Journal of Behavioral and Experimental Finance 42 (2024) 100910

Contents lists available at ScienceDirect

Journal of Behavioral and Experimental Finance

journal homepage: www.elsevier.com/locate/jbef

Lottery demand, weather and the cross-section of stock returns

Reza Bradrania *, Ya Gao

University of South Australia, Australia

ARTICLE INFO

JEL classification:
G11
G12
G14
G41

Keywords:
Lottery
Mood
Weather
Sentiment
Behavioral bias
Expected returns

ABSTRACT

We investigate the role that weather-induced mood plays in underperformance of lottery stocks. We hypothesize that during pleasant weather conditions, investors are more likely to become risk-taking and optimistic, and invest more in lottery stocks. This results in overpricing and lower expected return for these stocks. We use sky cloud cover as a proxy for the weather condition and show that the MAX effect mainly exists following sunny weather; however, there is less evidence for the MAX anomaly following cloudy weather. Our finding is robust to using alternative constructions of MAX and skewness measures as well as other proxies for weather condition such as rain, wind and temperature.

1. Introduction

Several studies provide evidence that investors exhibit strong preferences for lottery-like stocks, or stocks with positively skewed payoffs, and that they pay a premium for such stocks(see, e.g., Kumar, 2009; Bali, Cakici, and Whitelaw, 2011).[1] Brunnermeier, Gollier, and Parker (2007) follow Brunnermeier and Parker (2005) and develop a model based on optimal expectations theory, in which investors choose to believe that a large payoff is more likely than it really is; thus, biased beliefs drive the demand for lottery-like stocks, and their overvaluation and low subsequent returns. Also, in a theoretical model based on Tversky and Kahneman's (1992) cumulative prospect theory,[2] Barberis and Huang (2008) show that investors overweight the small probability of large payoffs of lottery-like stocks, which results in their overvaluation and low subsequent returns. In this model, the overweighing of tails does not indicate a bias in beliefs, but represents the common preference for the positively skewed payoffs of lottery-like stocks.

We argue that the preference of investors for lottery-like stocks has a stronger price impact during pleasant weather conditions. Studies in psychology have shown that pleasant or unpleasant weather conditions such as sunlight or lack of sunlight affect moods (e.g., Persinger, 1975; Cunningham, 1979; Schwarz and Clore, 1983; Parrott and Sabini, 1990; Scott, 2007), and that people who are in good (bad) moods can have more optimistic (pessimistic) evaluations of their environment, as moods influence their judgements and choices (e.g., Isen, Shalker, Clark, and Karp, 1978; Schwarz and Clore, 1983; Johnson and Tversky, 1983; Wright and Bower, 1992; Forgas, 1995). These findings have motivated financial economists to explore if weather, through mood, affects stock prices. Hirshleifer and Shumway (2003) and Saunders (1993) show that stock returns are significantly higher on sunny days than on cloudy days, indicating that stock prices are influenced by investor's emotional levels. Goetzmann and Zhu (2005) and Chang, Chen, Chou, and Lin (2008) show that weather affects stock trading activities through its effect on mood, and Bassi, Colacito, and Fulghieri (2013) provide evidence that sunshine can enhance risk-taking behavior through its effect on mood. Similarly, Cortés, Duchin, and Sosyura (2016) demonstrate that positive mood induced by sunshine increases the risk tolerance of loan officers, which in turn generates higher credit (or loan) approvals.

In this paper, we examine the role of weather in influencing the underperformance of lottery-like stocks. The intuition is that pleasant

* Corresponding author.
E-mail address: reza.bradrania@unisa.edu.au (R. Bradrania).

[1] Lottery-like stocks, or equivalently, lottery-type stocks refer to stocks with lottery-like payoffs (i.e., stocks with skewed distributions that have a small chance of extremely high payoffs).

[2] The cumulative prospect theory captures people's attitudes towards risk in experimental settings, indicating that people are risk-seeking when there is a small probability of winning a large prize (Tversky and Kahneman, 1992).

https://doi.org/10.1016/j.jbef.2024.100910
Received 13 September 2023; Received in revised form 1 March 2024; Accepted 1 March 2024
Available online 7 March 2024
2214-6350/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

weather conditions induce a good mood in investors, which biases their judgements and beliefs about the future prospects of lottery-like stocks. In the theoretical framework of Brunnermeier et al. (2007), to predict the pricing of positive skewness, an investor trades off the benefit of holding pleasurable (biased) beliefs against the cost that these beliefs may lead her or him to take unwise decisions. We conjecture that weather-induced good moods reinforce these pleasurable beliefs. This postulation is in line with the psychological evidence that people in good moods find positive material more salient and psychologically more available than negative material (e.g., Isen et al., 1978; Hirshleifer and Shumway, 2003), resulting in biased beliefs. The idea that weather conditions relate to investors' gambling preference is also consistent with the cumulative prospect theory (Tversky and Kahneman, 1992) as modeled in Barberis and Huang (2008); that is, the weather-induced good moods increase investors' risk-taking behavior and their overweighting of the small chance of high payoffs of lottery stocks.

To our knowledge, this study is the first to investigate how weather plays a role in lottery demand and the performance of lottery-like stocks. Recently, lottery demand has attracted the attention of the researchers in finance studying the existence and potential driving factors as well as the impact of lottery-related anomalies on financial market outcomes. Also, a large body of studies provide empirical evidence that mood influences financial decision making. We extend the research concerning the underlying drivers of the underperformance of lottery-type stocks by relating mood induced by weather conditions to gambling preference and provides some guidance on investment decisions depending on weather conditions.[3] This research is important as weather has been shown to have a significant effect on human moods and behavior. According to Persinger and Levesque (1983), weather conditions can help explain about 40% of daily variation in mood, and sunshine, as the most robust weather variable, can produce a large and monotonic effect on human moods (Persinger and Levesque, 1983; Howarth and Hoffman, 1984; Rind, 1996). In this paper, we relate weather, as an important factor that impacts human moods, to lottery demands which influence stock returns.

We aim to examine whether the underperformance of lottery-like stocks is stronger following good (or pleasant) weather conditions compared to bad (or unpleasant) weather conditions. To do so, in our tests we take the maximum daily return (MAX) over the last month, suggested by Bali et al. (2011) as the main proxy for lottery features of a stock. MAX has widely been used as a measure for positive skewness and lottery demand that captures investors' tendency to invest in lottery-like stocks (e.g., Fong and Toh, 2014; Bali, Brown, Murray, and Tang, 2017; Lin and Liu, 2018; Cheon and Lee, 2018; Byun et al., 2020), and underperformance of these stocks is commonly known as the MAX effect or MAX anomaly.[4] We adopt the deseasonalized sky cloud cover (DSKC) in the headquarter metropolitan city of a firm as the main proxy for the weather affecting the behavior of investors submitting orders in the stock market. Cloud cover measures sunshine, which is the most important weather variable shown to affect human moods (Persinger and Levesque, 1983; Howarth and Hoffman, 1984; Rind, 1996) and used in previous studies (e.g., Saunders, 1993; Hirshleifer and Shumway, 2003; Goetzmann, Kim, Kumar, and Wang, 2015; Cortés et al., 2016; Chhaochharia, Kim, Korniotis, and Kumar, 2019). We also use wind, rain and temperature as alternative proxies to identify relatively pleasant and unpleasant weather conditions. Prior research shows that rainy or windy weather, similar to cloudy weather, triggers negative mood states (e.g., Dehaan, Madsen, and Piotroski, 2017; Jiang, Norris, and Sun, 2021), and that higher temperature has a positive impact on mood (e.g., Cunnigham 1979; Howarth and Hoffman, 1984; Hu and Lee, 2020) and leads to risk-taking behavior (Keller et al., 2005).

We focus on Nasdaq stocks, since previous research on bias toward local trading[5] suggests that trading in these stocks is mostly localized (e.g., Loughran and Schultz, 2004).[6] We use portfolios of stocks of firms headquartered in metropolitan cities that are geographically disperse and span regions from the east to west coasts and north to south of the United States. As the initial step, we test and show that there is a MAX effect across our sample. This is important since our sample from Nasdaq is different from the samples used in previous research that has studied and documented lottery-related anomalies in the US market, which usually includes the common stocks listed on the American Stock Exchange (AMEX) and New York Stock Exchange (NYSE) as well as Nasdaq. Then we examine whether the MAX effect is stronger following weather-induced optimistic mood conditions than weather-induced pessimistic mood conditions.[7] We empirically test this hypothesis in several ways. First, our univariate portfolio analysis based on various thresholds of DSKC to define relatively sunny and cloudy weather conditions shows that, for both value- and equal-weighted portfolios, the difference between returns on the portfolios with the high and low MAX, is negative and significant after relatively sunnier months, but it is not statistically distinguishable from zero following relatively cloudy months. Second, we divide periods into two groups of relatively sunny and cloudy months for each metropolitan city and run monthly Fama and MacBeth (1973) cross-sectional regressions across all stocks for each subsample. The results show that the MAX effect is stronger after relatively sunny months compared to cloudy months. Our cross-sectional analysis is robust when we use DSKC and the interaction term between MAX and a dummy variable based on DSKC as control variables and run a single regression model instead of running cross-sectional analysis across two subsamples of sunny and cloudy months. Third, we use rain, wind and temperature as proxies for weather conditions and show that the MAX effect is more prevalent after months with relatively pleasant weather (less rain, less wind or higher temperature) than those with unpleasant weather (more rain, more wind or lower temperature).

The results are robust to several alternative tests such as using alternative constructions of MAX and skewness measures as proxies for lottery demand, and running portfolio analysis to detect the MAX effect when we use the component of MAX that is orthogonal to DSKC as the sort variable. In addition, we replace DSKC with SKC, which is not deseasonalized, as the weather-induced mood proxy in both portfolio analysis and the cross-sectional regressions. We also use deseasonalized rain, wind, and temperature variables as the proxies for weather-

[3] Prior research documents investor sentiment (Fong and Toh, 2014), levels of mispricing (Zhong and Gray, 2016), reference-dependent preference (An, Wang, Wang, and Yu, 2020), an upper bound for price named the 52-week high (Byun, Goh, and Kim, 2020), individualism (Cheon and Lee, 2018), individual investors' stock preference (Lin and Liu, 2018), attention and social interaction (Bali, Hirshleifer, Peng, and Tang, 2019), and property crime (Gao and Bradrania, 2024) as drivers of the lottery-related anomalies.

[4] Bali et al. (2011) first introduce this phenomenon in the US stock market. Following Bali et al. (2011), many studies explore the existence of the MAX effect in other stock markets such as the European market (Annaert, De Ceuster, and Verstegen, 2013, Walkshäusl, 2014), the South Korean market (Nartea, Wu, and Liu, 2014), the Australian market (Zhong and Gray, 2016; Bradrania and Veron, 2023), the Canadian market (Aboulamer and Kryzanowski, 2016), the Chinese market (Nartea, Kong, and Wu, 2017), the Brazilian market (Berggrun, Cardona, and Lizarzaburu, 2019), the global market (Cheon and Lee, 2018), and nine advanced emerging markets (Seif, Docherty, and Shamsuddin, 2018). We use MAX effect and MAX anomaly interchangeably hereafter.

[5] There is strong evidence that investors hold and trade significantly more in local firms (Coval and Moskowitz, 1999; Coval and Moskowitz, 2001; Grinblatt and Keloharju, 2001; Huberman, 2001; Zhu, 2002; Ivković and Weisbenner, 2005; Seasholes and Zhu, 2010).

[6] Loughran Schultz (2003) and Anand et al. (2011) show that Nasdaq market makers exhibit a strong preference for stocks of companies that are located near them. Loughran and Schultz (2004) is motived by these studies and provide evidence that trading behavior in Nasdaq is localized.

[7] Throughout this paper, we use optimistic (pessimistic), positive (negative) or good (bad) mood conditions interchangeably.

induced mood in the cross-sectional regressions. These robustness tests are discussed in detail in the Internet Appendix.

Our findings provide evidence that the weather effect, which refers to the impact of weather on mood, plays a role in lottery demand. Previous literature documents that an optimistic (pessimistic) mood induced by pleasant (unpleasant) weather conditions can yield a more optimistic (pessimistic) evaluation or expectation of the situation faced (Hirshleifer and Shumway, 2003; Cortés et al., 2016; Chhaochharia et al., 2019), encourage (discourage) risk-taking behavior (Bassi et al., 2013; Cortés et al., 2016), raise an increased (decreased) desire to trade and a propensity for buying (selling) stocks (Loughran and Schultz, 2004; Goetzmann and Zhu, 2005; Goetzmann et al., 2015; Chang et al., 2008), and have a positive (negative) influence on stock returns (Saunders, 1993; Hirshleifer and Shumway, 2003; Goetzmann and Zhu, 2005). We extend this literature by showing that weather-based indicators of mood impact the underperformance of lottery stocks. We show that the underperformance of lottery-like stocks is stronger following pleasant weather conditions than unpleasant weather conditions. Our findings are consistent with the findings of Chhaochharia et al. (2019) and Schwarz, (1990), which suggest that the effects of good (optimistic) and bad (pessimistic) moods on returns is asymmetric in that biases caused by good mood are stronger than those caused by bad mood. In a relevant study, Loughran and Schultz (2004) confirm localized trading of Nasdaq stocks and show that firms headquartered in cities experiencing a blizzard tend to exhibit a decrease in trading volume, indicating that weather affects the trading volume of stocks. However, they find little evidence that cloudy weather has an impact on stock returns. We also focus on Nasdaq stocks and provide evidence that weather affects the trading activities of lottery stocks. Unlike Loughran and Schultz (2004), we show that local weather impacts the stock returns and especially for the lottery-type stocks.

Fong and Toh (2014) provide evidence that investor sentiment, proxied by Baker and Wurgler's (2006) measure of economic sentiment,[8] impacts the MAX anomaly.[9] Our paper is different since we focus on mood rather than sentiment as a driver of lottery demand. While 'sentiment' and 'mood' are often used interchangeably in the extant research, they are different concepts from a theoretical perspective. Financial or economical sentiment refers to the general sense of investors' sentiments toward the economy or financial markets, and is a concept of behavioral finance, whereas mood is a concept of psychology and refers to the emotional states of investors (Rapp, 2019). Also, while sentiment arises due to biased cognitive processes of investors (Baker and Wurgler, 2007), mood arises due to biological processes (Rosenthal et al., 1984). Moreover, our proxy for mood is an exogenous environmental factor, which is different from the sentiment proxies constructed from observable economic activities such as the Baker and Wurgler sentiment index. In addition, unlike sentiment proxies based on the national measures of economic sentiment that assume market-wide homogeneity, our proxy for investor mood provides rich cross-sectional variation across different geographic regions.

In sum, our study provides strong evidence on how weather, through its effect on mood, affects investors' financial decisions and financial market outcomes, which have implications for investors' portfolio choice and financial markets. For instance, our findings suggest that investors should consider reducing their investments in lottery-type stocks when the weather conditions are pleasant, since these stocks are overpriced during such periods. In addition, while our study focuses on the relationship between weather-induced mood and lottery-demands, which influence stock returns, our findings suggest that weather may impact other high-risk investment vehicles such as CFDs and binary options. We leave examining this possibility for further research.

The rest of the paper is organized as follows. In Section 2, we present the relevant data sources. Section 3 discusses the lottery demand and stock returns. In Section 4, we show the empirical results for the association of weather-induced mood and lottery demand, and Section 5 concludes.

2. Data sources and sample

In this study, we include all common stocks traded on the Nasdaq from January 1983 to December 2019. We focus on Nasdaq stocks since previous research suggests that trading in Nasdaq stocks is localized (Loughran and Schultz, 2004). We follow Pirinsky and Wang (2006) and Loughran and Schultz (2004) and define a firm's location as the location of its headquarters.

To identify locations, we first obtain state, county and city, and ZIP codes of the firms' headquarters in the US for each year from Compustat. Using the ZIP code, we then merge the sample of firms with the 5-Digit ZIP Code Database[10] which includes Metropolitan Statistical Areas (MSAs) defined by the Office of Management and Budget (OMB). We follow Loughran and Schultz (2004) and define a firm's location as the MSA of its headquarters. OMB defines a MSA as a core area with at least one urbanized section with a population of 50,000 or more, plus adjacent communities that have a high degree of social and economic integration with that core. MSAs involve one or more counties, and some MSAs may contain counties from several states. For example, Boston MSA includes New Hampshire and Massachusetts. Throughout this paper, we use MSAs, areas, regions, and metropolitan cities interchangeably.

The sample includes only firms based in 10 MSAs with the largest number of Nasdaq-listed companies. They are New York, Boston, San Jose, Los Angeles, Chicago, Dallas, Houston, Washington D.C., Philadelphia, and San Francisco. These metropolitan cities have a total population of more than 56 million[11] and offer a good representation of US urban residents, leading to a fair sample of US investors. In addition, these cities' metropolitan areas are situated across the US from the east to west coasts, which have quite different weather conditions. Panel A of Table 1 presents the details of the metropolitan cities in our study as well as the number of firms within each metropolitan city during the sample period.

We follow Loughran and Schultz (2004) and use the local weather of metropolitan cities in which Nasdaq-listed companies are headquartered as an investor mood proxy. We source the weather data from the Integrated Surface Database (ISD-lite), which is provided by the National Oceanic and Atmospheric Administration (NOAA).[12] We collect the hourly sky cloud cover (SKC), wind speed rate, liquid precipitation depth dimension, and air temperature from this dataset. The hourly SKC measures the total celestial dome covered by clouds, ranging from 0 (clear or the least cloudy weather) to 8 (overcast or the cloudiest weather). The hourly wind speed is the rate of horizontal travel of air past a fixed point measured by the units of meters per second. The depth of liquid precipitation, rain, is measured by the units of millimeters over a one-hour accumulation period. The hourly temperature of the air is

[8] The Baker and Wurgler sentiment index generally reflects investors' optimism or pessimism about the stock market. It consists of six underlying sentiment proxies, which involve closed-end fund discount, the number and the first-day returns of IPOs, NYSE turnover, the equity shares in total new issues, and the dividend premium.

[9] Stambaugh, Yu, and Yuan (2012) also use Baker and Wurgler's (2006) sentiment index as the proxy for investor sentiment and find that the lottery-related anomalies related to failure probability (Campbell, Hilscher, and Szilagyi, 2008) and bankruptcy probability (Ohlson, 1980) are more pronounced following high sentiment periods.

[10] https://zipcodedownload.com/

[11] https://www.census.gov/data/tables/time-series/demo/popest/2010s-total-metro-and-micro-statistical-areas.html

[12] https://www.ncei.noaa.gov/pub/data/noaa/isd-lite/

Table 1
Summary statistics. Panel A reports the distribution of the number of firms across metropolitan cities (MSAs). Panel B presents the summary statistics of weather variables across MSAs: deseasonalized sky cloud cover (DSKC), sky cloud cover (SKC), rain (RAIN, scaled by 10), wind (WIND, scaled by 10) and maximum temperature (TEMP, scaled by 10). Panel C provides summary statistics for the main stock characteristic variables used in the analysis: the maximum daily return over a month (MAX) in percent, the market beta (BETA), the total skewness (TSKEW), the idiosyncratic skewness (ISKEW), the Amihud (2002) measure of illiquidity scaled by 10^6 (ILLIQ), the natural log of market capitalization in millions of dollars (SIZE), momentum in percent (MOM), short-term reversal in percent (REV), and the book-to-market ratio (BTM). All variables are defined in the Appendix. The sample includes Nasdaq-listed firms headquartered in the 10 largest US MSAs, and the sample period is from January 1983 to December 2019.

Panel A: The distribution of the number of firms across MSAs

Metropolitan City	MSA Code	MSA Name	Number of Nasdaq-Listed Firms
1 New York	5600	New York, NY	726
2 Boston	1120	Boston, MA/NH	628
3 San Jose	7400	San Jose, CA	551
4 Los Angeles	4480	Los Angeles/Long Beach, CA	531
5 Chicago	1600	Chicago, IL	419
6 Dallas	1920	Dallas, TX	365
7 Houston	3360	Houston, TX	354
8 Washington, D.C.	8840	Washington, DC/MD/VA/WV	353
9 Philadelphia	6160	Philadelphia, PA/NJ	346
10 San Francisco	7360	San Francisco, CA	336
Total	-	-	4609

Continued on next page…
…Continued

Panel B: Summary statistics of weather variables across the MSAs

Variables	Mean	Standard Deviation	25th percentile	Median	75th percentile
DSKC	0.0021	0.9196	–0.5378	0.0717	0.6048
SKC	4.3027	1.1307	3.5836	4.4798	5.1261
RAIN	0.9515	1.3782	0.2370	0.7182	1.3519
WIND	35.3415	12.2166	26.4633	36.1730	44.0909
TEMP	160.4303	81.5436	108.3667	160.1452	222.2759

Panel C: Summary statistics of stock characteristic variables

Variables	Mean	Standard Deviation	25th percentile	Median	75th percentile
MAX	9.69	11.76	4.08	6.86	11.54
BETA	0.93	5.44	–0.17	0.81	2.03
TSKEW	0.25	1.13	–0.27	0.22	0.77
ISKEW	0.23	1.07	–0.28	0.20	0.75
ILLIQ	5.19	19.72	0.01	0.11	1.28
SIZE	4.68	2.02	3.30	4.61	5.93
MOM	14.15	86.77	–30.18	0.40	35.22
REV	1.16	21.63	–8.63	0.00	8.40
BTM	0.67	0.81	0.25	0.48	0.85

measured by the units of degrees Celsius. The hourly measures of wind, rain and temperature are all scaled by 10. There is usually more than one weather reporting station near a city's metropolitan area. To be consistent for all locations, we collect the hourly observations of sky cloud cover, wind, rain and temperature reported at each MSA's major airport[13] from January 1983 to December 2019. We require this data to be available for a minimum of three hours each day and a minimum of 11 days each month,[14] and we follow Hirshleifer and Shumway (2003) and only retain observations between the hours of 6 AM to 4 PM, which is the period of time when investors are likely to observe outdoor weather conditions. For each MSA, we compute the daily SKC as the average of hourly SKC values of each day and then calculate the monthly SKC[15] as the average of daily estimates over the month (e.g., Goetzmann and Zhu, 2005; Chhaochharia et al., 2019). We use a similar procedure to calculate monthly wind, rain and temperature. Previous research suggests that higher temperature induces a positive mood in investors (Cunnigham 1979; Howarth and Hoffman, 1984; Hu and Lee, 2020) and leads to risk-taking behavior (Keller et al., 2005). For each MSA, we follow Hu and Lee (2020) and define the temperature variable, as the monthly maximum temperature, calculated as the average of the daily maximum temperature levels over the month.[16] Also, prior literature suggests that SKC has a seasonality pattern (e.g., Hirshleifer and Shumway, 2003; Goetzmann and Zhu, 2005; Chhaochharia et al., 2019). We therefore follow Goetzmann and Zhu (2005) and compute monthly deseasonalized sky cloud cover (DSKC) for each MSA as the difference between current month SKC and average SKC for the same month over the entire sample period.

DSKC is the main proxy for investors' weather-induced mood in our analysis,[17] and we use rain (RAIN), wind (WIND) and maximum

[13] Loughran and Schultz (2004) also use the weather data reported at each MSA's major airport as the proxy for the local weather of MSAs.

[14] We follow Dehaan, Madsen, and Piotroski (2017) and Jiang, Norris, and Sun (2021) to set our cut off of 3hrs for the day. To calculate the daily cloud cover, wind, and rain for each weather station, Dehaan et al. (2017) and Jiang et al. (2021) require the cloud cover, wind, and rain data to be available for a minimum of 4 hours each day, and only retain observations between the hours of 6 am to 6 pm. This is 1/3 of the 12hrs time span between 6 am to 6 pm. However, we follow Hirshleifer and Shumway (2003) and only retain weather data from 6 am to 4 pm, which is the time span of 10 hours. So we set the cut off at 1/3 of 10 which is circa 3hrs. This cut off point allows us to keep 95.81% of hourly observation data and 98.78% of daily observation data in the final sample. Also, the summary statistics of non-missing hours observations suggest that the cut off point is reasonable and do not bias our findings.

[15] We construct monthly metrics of the weather data since we use monthly frequency in our pricing and related empirical tests, following the literature on empirical asset pricing that uses monthly frequency data to avoid microstructure bias.

[16] The daily maximum temperature is based on the value of maximum hourly temperature of each day.

[17] The results remain robust when we use SKC as the proxy for weather-induced mood, shown in the Internet Appendix.

temperature (TEMP) as a proxy for weather-induced mood in our robustness analysis. Panel B of Table 1 shows the summary statistics of weather variables in our sample. These include the mean, standard deviation, 25th percentile, median, and 75th percentile of the variables. The mean (median) of DSKC among the metropolitan cities is 0.0021 (0.0717) per month during the sample period, and the mean (median) of sky cloud cover (SKC) among the MSAs is 4.3027 (4.4798) per month during the sample period.

We collect transaction data (price, returns, volume and number of shares outstanding) from the Center for Research in Security Prices (CRSP). We use daily returns to calculate the maximum daily stock returns (MAX) for each stock in each month as well as the market beta (BETA) and skewness measures (TSKEW and ISKEW). The daily returns and volume data are used to calculate the Amihud (2002) measure of illiquidity (ILLIQ). The proxy for market return is the CRSP daily value-weighted market index. We use share price and shares outstanding from monthly data to compute market capitalization (MKTCAP). We use monthly returns to calculate proxies for intermediate-term momentum (MOM) and short-term reversals (REV). Finally, we collect equity book value of firms from Compustat to compute the book-to-market ratios (BTM) of individual firms. Panel C of Table 1 reports the summary statistics of these main stock characteristic variables.

There are on average 4609 firms in our sample period after merging transaction and weather data. All variables constructed using datasets mentioned in this section are defined in detail in the Appendix and are discussed as they are used in the analysis.

3. Lottery demand and stock expected returns

In this section, we show that there is a MAX effect in our sample of stocks. Each month, we sort all stocks in our sample into deciles based on the maximum daily return (MAX) during the previous month, with each decile having an equal number of stocks. Table 2 presents the average monthly portfolio raw returns and alphas for the equal- and value-weighted deciles and for the difference between decile 10 and decile 1. The alphas are risk-adjusted returns relative to the CAPM, Fama and French (1993) three-factor model (FF3), and Fama and French (1993) and Carhart (1997) four-factor model (FFC4).[18] The numbers in parentheses are *t*-statistics, adjusted following the Newey-West (1987) method using six lags.

The results show that the lottery-like portfolio of stocks (decile 10) significantly underperforms the non-lottery-like portfolio (decile 1). The average raw return for the value-weighted zero-cost portfolio, which is long the high MAX portfolio and short the low MAX portfolio, is −1.06% per month (*t*-statistic = −2.31).

The alphas for value-weighted zero-cost portfolios are stronger. The CAPM, FF3, and FFC4 alphas are −1.62% (*t*-statistic = −3.96), −1.33% (*t*-statistic = −3.64), and −1.10% (*t*-statistic = −2.64) per month, respectively. Similar to Bali et al. (2011), our results for equal-weighted portfolios are less significant both economically and statistically. The equal-weighted average raw return of the high-low MAX portfolio is −0.54% per month (*t*-statistic = −1.33). However, after adjusting for risk, the difference between returns of the high and low MAX portfolios becomes stronger. For example, CAPM and FF3 alphas are −1.10% (*t*-statistic = −3.00) and −0.84% (*t*-statistic = −2.86) per month, respectively.

Overall, the results in Table 2 show that there is a negative relation between extreme positive daily return (MAX) in the prior month and the cross-section of expected returns in our sample.

To better understand the characteristics of the lottery and non-lottery stocks, we report the summary statistics of each MAX decile portfolio in Table 3. For each decile, we present the monthly averages of the median values of various characteristics of the stocks in our sample.[19] The median of MAX increases from 1.78% to 23.88% moving from the low MAX to the high MAX deciles, suggesting that the distribution of maximum daily returns is positively skewed by some stocks that have extremely high returns. PRICE and MKTCAP decrease as MAX increases across the deciles, indicating that the higher MAX portfolios are dominated by firms with lower prices and smaller market capitalizations. The median of BETA increases from the low MAX to the high MAX deciles, suggesting that stocks with higher MAX are likely to be exposed to market risk. The median BTM ratio seems to be slightly changed across the ten deciles and follow a U-shape pattern from the low MAX to the high MAX deciles. The median of REV increases from the low MAX to the high MAX deciles monotonically, indicating that stocks with a high maximum daily return also have a high return in that month. The negative values of MOM for the higher MAX decile portfolios are also consistent with the fact that lottery-type stocks tend to have lower average returns. Illiquidity decreases from decile 1–2, then significantly increases for the higher deciles, which indicates that higher MAX portfolios contain more illiquid stocks.

In the next sections, we empirically test whether the investors' weather-induced mood plays a significant role in influencing MAX effect and drives this anomaly.

4. Weather-induced mood and lottery demand

4.1. Univariate portfolio analysis

To test if weather conditions have an impact on the MAX anomaly, we classify periods into two groups of sunny and cloudy months based on the firm's DSKC, which measures the sunny and cloudy weather conditions. More specifically, for each MSA we create a ranking of DSKC over time.[20] We classify months in which the one-month lagged DSKC is in the lowest (or bottom) and highest (or top) threshold for DSKC as Sunny and Cloudy, respectively. We consider the 10th, 20th, 25th, 40th, 50th percentiles as well as mean of DSKC as alternative thresholds for DSKC. For each subsample and in each month, we sort sample stocks into deciles based on MAX. Table 4 provides the time-series averages of the cross-sectional means of returns for deciles following Sunny and Cloudy months.[21] It also includes the return differences between the high and low MAX deciles. Panels A and B present the results for value- and equal-weighted deciles, respectively. The *t*-statistics are adjusted for heteroscedasticity and autocorrelation using the Newey-West (1987) method

[18] The FFC4 alpha is the estimated intercept coefficient from a regression of the excess portfolio return on excess return of the market portfolio (MKTRF), a size factor (SMB), a book-to-market factor (HML) and a momentum factor (UMD), following Fama and French (1993) and Carhart (1997). We use the same approach to calculate FF3 and CAPM alphas by using three factors (MKTRF, SMB and HML) and the market factor (MKTRF) as risk factors in regressions, respectively. MKTRF, small minus big (SMB), high minus low (HML), and winner minus loser (UMD) are described in and obtained from Kenneth French's data library (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/).

[19] We present the median statistics, since it is a robust measure of the center of the distribution and less sensitive to outliers than the mean. In unreported results, we find that the results from the mean statistics are qualitatively similar to those obtained from the median statistics in unreported tests.

[20] The reason that we use the thresholds of DSKC within each MSA is that investors' mood based on weather is determined by variation of weather in the same region where they live. Since some regions are systematically exposed to extreme (cloudier or sunnier) weather, the relation between weather and mood could vary geographically (Chhaochharia et al., 2019). We also make a ranking of DSKC across all MSAs where firms are headquartered in each month to classify our sample firms into subsamples based on sunny and cloudy weather conditions in Table IA6 of the Internet Appendix, and find the results are qualitatively similar.

[21] For brevity, we do not report the results of the risk-adjusted returns for the MAX decile portfolios in Table 4.

Table 2
MAX-sorted portfolios Decile portfolios are formed every month from January 1983 to December 2019 by sorting stocks based on the maximum daily return (MAX) over the previous month. Decile 1 (10) is the portfolio of stocks with the lowest (highest) MAX. 10−1 is the zero-cost portfolio, which is long the decile 10 portfolio (High MAX) and short the decile 1 portfolio (Low MAX). The table reports the average monthly returns as well as the CAPM alphas, Fama-French three-factor alphas (FF3 alpha) and Fama-French-Carhart four-factor alphas (FFC4 alpha) on the value-weighted and equal-weighted portfolios. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs, and returns and alphas are in percent. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

	Value-Weighted Portfolios				Equal-Weighted Portfolios			
Decile	Average return	CAPM alpha	FF3 alpha	FFC4 alpha	Average return	CAPM alpha	FF3 alpha	FFC4 alpha
1 (Low MAX)	1.45	0.52	0.53	0.52	1.25	0.47	0.42	0.47
2	1.43	0.33	0.46	0.42	1.50	0.52	0.53	0.57
3	1.50	0.26	0.44	0.49	1.50	0.39	0.43	0.54
4	1.38	0.11	0.32	0.39	1.26	0.09	0.19	0.32
5	1.22	–0.17	0.13	0.19	1.24	0.01	0.14	0.32
6	0.90	–0.52	–0.23	–0.15	1.14	–0.10	0.07	0.24
7	0.96	–0.52	–0.26	–0.09	1.31	0.00	0.18	0.44
8	0.59	–0.86	–0.56	–0.43	0.89	–0.44	–0.27	0.03
9	0.65	–0.81	–0.44	–0.36	0.99	–0.36	–0.20	0.12
10 (High MAX)	0.39	–1.10	–0.80	–0.58	0.70	–0.63	–0.42	–0.03
10–1	–1.06	–1.62	–1.33	–1.10	–0.54	–1.10	–0.84	–0.50
	(–2.31)	(–3.96)	(–3.64)	(–2.64)	(–1.33)	(–3.00)	(–2.86)	(–1.73)

Table 3
Summary statistics for MAX-sorted portfolios. Decile portfolios are formed every month from January 1983 to December 2019 by sorting stocks based on the maximum daily return (MAX) over the previous month. For each decile, the table presents the monthly time-series average of the cross-sectional medians of various characteristics in each month. The characteristic variables are the maximum daily return in percent (MAX), the stock price in dollars (PRICE), the market capitalization in millions of dollars (MKTCAP), the market beta (BETA), the book-to-market ratio (BTM), the short-term reversal in percent (REV), the momentum in percent (MOM), and the Amihud (2002) measure of illiquidity scaled by 10^6 (ILLIQ). All variables are defined in the Appendix.

Decile	MAX	PRICE ($)	MKTCAP ($ Millions)	BETA	BTM	REV	MOM	ILLIQ (10^6)
1 (Low MAX)	1.78	15.14	334.29	0.30	0.6447	–1.88	7.72	0.2754
2	3.32	17.93	398.67	0.61	0.5471	–1.79	12.44	0.1428
3	4.34	15.55	341.92	0.77	0.5043	–1.47	10.54	0.1622
4	5.30	13.43	283.83	0.90	0.4681	–1.09	8.48	0.1887
5	6.33	11.33	225.23	0.99	0.4700	–0.63	6.82	0.2355
6	7.51	9.62	186.69	1.09	0.4621	0.03	3.11	0.3114
7	8.96	8.15	158.06	1.16	0.4589	0.90	0.17	0.4342
8	10.97	6.52	127.70	1.17	0.4735	2.13	–5.11	0.6970
9	14.39	4.96	97.54	1.25	0.4972	4.37	–11.91	1.2801
10 (High MAX)	23.88	3.22	60.81	1.19	0.5589	11.13	–26.78	4.2772

with six lags.

The results for value-weighted portfolios in Panel A show that the MAX effect is more economically and significantly negative following relatively pleasant weather conditions (sunny months with low DSKC), however, it tends to be insignificant following relatively unpleasant weather conditions (cloudy months with high DSKC). For example, using the 10th percentile threshold of DSKC, the raw return for the high-low MAX portfolio in the bottom 10th percentile of DSKC (Sunny group) is −1.83% per month with a *t*-statistic of −2.90; however, the corresponding raw return in the top 10th percentile of DSKC (Cloudy group) is 0.04% per month with a *t*-statistic of 0.05. We find similar results for those risk-adjusted returns. For example, the Fama-French-Carhart four-factor (FFC4) alpha of the high-low MAX portfolio in the bottom 10th DSKC percentile is −1.98% per month with a *t*-statistic of −3.11, which is more negative and statistically significant compared with the insignificant corresponding return in the top 10th percentile of DSKC (−0.19% per month with a *t*-statistic of −0.22). The results for equal-weighted portfolios in Panel B provides similar evidence that the MAX effect is stronger following weather-induced positive mood conditions (sunny weather with low DSKC) than weather-induced negative mood conditions (cloudy weather with high DSKC). We find similar results for returns after adjusting for risk factors.

In summary, the results in Table 4 support our argument that the MAX effect is impacted by weather conditions. The MAX effect becomes stronger following weather-induced optimistic mood conditions (sunny weather) than weather-induced pessimistic mood conditions (cloudy weather).

4.2. Cross-sectional regression analysis

In this section, we run Fama-MacBeth (1973) cross-sectional regressions across firms and test whether the MAX effect is stronger (weaker) following months associated with weather-induced optimistic (pessimistic) mood conditions, measured by sunny (cloudy) months with low (high) DSKC. The cross-sectional regression approach allows us to simultaneously control for other variables when assessing the relation between MAX and future stock returns under different weather conditions.

To isolate the effect of different weather conditions on the MAX effect, we run the regressions under sunny and cloudy weather conditions separately. Similar to our approach in the univariate analysis to separate periods into two groups of sunny and cloudy months, for each MSA we create a ranking of DSKC over time and classify months in which the one-month lagged DSKC is in the lowest and highest threshold for DSKC as Sunny and Cloudy, respectively. We consider the 10th, 20th, 25th, 40th and 50th percentiles as well as mean of DSKC as alternative thresholds for DSKC. For each sunny and cloudy weather group, we run the following monthly cross-sectional regression:

$$R_{i,t} = \lambda_{0,t-1} + \lambda_{1,t-1} MAX_{i,t-1} + \boldsymbol{\Lambda}_{t-1} \boldsymbol{Z}_{i,t-1} + \mathscr{E}_{i,t} \tag{1}$$

where $R_{i,t}$ is the realized return (in percent) on stock i in month t, $MAX_{i,t-1}$ is the maximum daily return for stock i in month $t-1$ and $\boldsymbol{Z}_{i,t-1}$ is a vector containing the control variables. We follow Bali et al. (2011) and control for variables that are widely considered as determinants of stock returns in the cross-section and include market beta (BETA),

Table 4
Weather-induced mood and MAX effect: univariate analysis. The sample stocks are classified into two subsample groups (Sunny or Cloudy) based on the firm's previous month DSKC ranking. The ranking of DSKC is created within each MSA where firms are headquartered and over time. The thresholds (DSKC% Split) used in ranking include the 10th, 20th, 25th, 40th, 50th percentiles as well as mean of DSKC. If a firm's monthly DSKC is in the bottom (top) threshold of DSKC, the firm-month is classified into the Sunny (Cloudy) subsample. For each subsample, decile portfolios are formed every month from January 1983 to December 2019 by sorting stocks based on the maximum daily return (MAX) over the previous month. Decile 1 (10) is the portfolio of stocks with the lowest (highest) MAX. 10−1 is the zero-cost portfolio, which is long the decile 10 portfolio (High MAX) and short the decile 1 portfolio (Low MAX). The table reports the value-weighted (Panel A) and equal-weighted (Panel B) average monthly returns as well as the CAPM alphas, Fama-French three-factor alphas (FF3 alpha) and Fama-French-Carhart four-factor alphas (FFC4 alpha) on the portfolios for Sunny and Cloudy subsamples. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs, and returns and alphas are in percent. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

Panel A: Value-Weighted Returns												
Decile	DSKC% Split											
	10th		20th		25th		40th		50th		Mean	
	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy
1 (Low MAX)	1.03	1.61	1.11	1.22	1.12	1.11	1.25	1.00	1.39	1.08	1.38	1.08
2	0.81	2.31	0.94	1.34	1.11	1.30	1.39	1.35	1.44	1.44	1.33	1.40
3	0.93	2.06	0.85	1.17	1.16	0.88	1.21	1.05	1.22	1.32	1.24	1.29
4	0.57	1.35	0.84	1.36	1.19	1.20	1.23	1.23	1.22	1.28	1.18	1.32
5	0.78	1.59	1.45	1.17	1.11	0.92	0.95	1.07	1.07	1.23	0.95	1.23
6	0.55	1.26	0.80	0.93	0.91	0.79	0.88	1.05	1.08	1.12	1.01	1.08
7	0.66	1.63	1.28	1.35	1.06	1.02	0.97	1.00	0.97	0.99	0.95	1.01
Continued on next page												
Continued												
8	0.64	1.65	0.80	1.05	0.50	0.42	0.61	0.88	0.71	0.94	0.69	0.82
9	0.61	0.62	0.64	0.84	0.16	0.43	0.63	0.40	0.85	0.57	0.71	0.70
10 (High MAX)	–0.81	1.66	–0.16	1.30	–0.16	0.62	0.16	0.95	0.15	0.78	0.09	0.71
10–1	–1.83	0.04	–1.28	0.09	–1.27	–0.49	–1.08	–0.06	–1.24	-0.31	-1.29	-0.37
	(–2.90)	(0.05)	(–2.28)	(0.14)	(–2.43)	(–0.82)	(–2.12)	(–0.13)	(–2.50)	(-0.69)	(-2.60)	(-0.83)
CAPM alpha	–2.15	–0.75	–1.69	–0.54	–1.75	–0.96	–1.50	–0.49	–1.74	-0.82	-1.73	-0.87
	(–3.42)	(–0.80)	(–3.16)	(–0.85)	(–3.48)	(–1.70)	(–3.37)	(–1.14)	(–4.07)	(-1.97)	(-3.97)	(-2.07)
FF3 alpha	–1.97	–0.34	–1.58	–0.11	–1.50	–0.68	–1.34	–0.17	–1.47	-0.52	-1.48	-0.58
	(–3.28)	(–0.41)	(–3.15)	(–0.19)	(–3.56)	(–1.29)	(–3.46)	(–0.46)	(–3.97)	(-1.47)	(-3.99)	(-1.59)
FFC4 alpha	–1.98	–0.19	–1.37	0.14	–1.12	–0.37	–1.04	0.17	–1.10	-0.16	-1.13	-0.22
	(–3.11)	(–0.22)	(–2.76)	(0.23)	(–2.68)	(–0.68)	(–2.63)	(0.40)	(–2.88)	(-0.44)	(-2.92)	(-0.58)
Panel B: Equal-Weighted Returns												
Decile	DSKC% Split											
	10th		20th		25th		40th		50th		Mean	
	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy	Sunny	Cloudy
1 (Low MAX)	1.08	1.67	1.13	1.27	1.16	1.14	1.32	1.03	1.46	1.12	1.45	1.11
2	0.78	2.30	0.94	1.34	1.10	1.31	1.40	1.36	1.46	1.44	1.34	1.41
3	0.96	2.11	0.88	1.20	1.18	0.90	1.24	1.05	1.23	1.34	1.26	1.30
4	0.58	1.29	0.84	1.36	1.22	1.23	1.25	1.24	1.24	1.28	1.20	1.32
5	0.76	1.62	1.46	1.16	1.12	0.93	1.00	1.10	1.11	1.25	0.99	1.26
6	0.59	1.30	0.83	1.01	0.94	0.86	0.92	1.12	1.11	1.17	1.04	1.13
7	0.66	1.68	1.32	1.42	1.13	1.09	1.05	1.07	1.04	1.05	1.02	1.06
8	0.72	1.70	0.89	1.11	0.55	0.49	0.65	0.95	0.76	1.02	0.74	0.89
9	0.71	0.68	0.68	0.89	0.18	0.49	0.66	0.48	0.92	0.63	0.75	0.77
10 (High MAX)	-0.75	1.80	-0.07	1.46	-0.06	0.76	0.32	1.12	0.27	0.92	0.24	0.85
Continued on next page												
Continued												
10–1	–1.83	0.13	–1.20	0.18	–1.22	–0.39	–1.00	0.08	–1.20	–0.19	–1.21	–0.26
	(–2.79)	(0.13)	(–2.06)	(0.27)	(–2.25)	(–0.63)	(–1.88)	(0.18)	(–2.34)	(–0.42)	(–2.37)	(–0.57)
CAPM alpha	–2.16	-0.67	–1.64	–0.46	–1.71	–0.87	–1.42	–0.36	–1.70	–0.71	–1.66	–0.77
	(–3.27)	(-0.68)	(–2.93)	(–0.69)	(–3.29)	(–1.48)	(–3.07)	(–0.80)	(–3.85)	(–1.64)	(–3.68)	(–1.76)
FF3 alpha	–1.98	–0.26	–1.51	–0.02	–1.45	–0.57	–1.26	–0.03	–1.44	–0.41	–1.41	–0.48
	(–3.12)	(–0.29)	(–2.90)	(-0.04)	(–3.32)	(–1.04)	(–3.09)	(–0.08)	(–3.68)	(–1.10)	(–3.60)	(–1.24)
FFC4 alpha	–1.97	–0.08	–1.28	0.26	–1.03	–0.24	–0.94	0.33	–1.04	–0.02	–1.04	–0.09
	(–2.94)	(–0.09)	(–2.47)	(0.40)	(–2.38)	(–0.42)	(–2.24)	(0.74)	(–2.58)	(–0.06)	(–2.51)	(–0.22)

natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). Detailed information about the control variables is provided in the Appendix. Table 5 presents the time-series averages of slope coefficients from the cross-sectional regressions following sunny as well as cloudy weather conditions. We report the results for the univariate model and a multivariate model with all control variables. The Newey-West (1987) adjusted *t*-statistics using six lags are presented in parentheses.

The results show that consistent with the portfolio analyses, the MAX effect is stronger following sunny weather compared to cloudy weather, in both univariate and multivariate models and across various samples defined based on different thresholds for DSKC. The average coefficients of MAX are significant in the univariate test following the sunny weather conditions, ranging between −0.042 (*t*-statistic = −2.38) and −0.058 (*t*-statistic = −2.82), associated with DSKC thresholds of 20th and 10th, respectively. However, following the cloudy weather conditions, the average slope coefficient of MAX in univariate tests is less negative and statistically significant, and varies between −0.026 (*t*-statistic = −1.95) and 0.006 (*t*-statistic = 0.28) where the DSKC threshold is equal to mean or the 20th percentile of DSKC, respectively.

The MAX effect following sunny weather is even stronger when we add control variables in the multivariate setting compared to the univariate model. The coefficients on MAX become more negative with values ranging from −0.043 (*t*-statistic = −3.17) to −0.074 (*t*-statistic = −2.05) for DSKC thresholds equal to the median and 10th percentile of

Table 5

Fama-MacBeth regressions. The sample stocks are classified into two subsample groups (Sunny or Cloudy) based on the firm's previous month DSKC ranking. The ranking of DSKC is created within each MSA where firms are headquartered and over time. The thresholds (DSKC% Split) used in ranking include the 10th, 20th, 25th, 40th, 50th percentiles as well as mean of DSKC. If a firm's monthly DSKC is in the bottom (top) threshold of DSKC, the firm-month is classified into the Sunny (Cloudy) subsample. For each subsample and every month from January 1983 to December 2019, we run the Fama-MacBeth regression of the monthly stock returns (in percent) on one-month lagged independent variables that include the maximum daily return (MAX) and a set of control variables. The control variables include market beta (BETA), natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). All variables are defined in the Appendix. The table reports the time-series average of the slope coefficients from cross-sectional regressions over Sunny and Cloudy subsamples. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

Variables	DSKC% Split											
	10th				20th				25th			
	Sunny		Cloudy		Sunny		Cloudy		Sunny		Cloudy	
MAX	-0.058	-0.074	-0.019	-0.059	-0.042	-0.048	0.006	-0.028	-0.049	-0.050	-0.009	-0.042
	(-2.82)	(-2.05)	(-0.76)	(-1.55)	(-2.38)	(-1.79)	(0.28)	(-1.11)	(-3.09)	(-2.48)	(-0.52)	(-1.96)
BETA		-0.060		-0.062		-0.205		0.106		-0.228		0.112
		(-0.51)		(-0.72)		(-1.58)		(1.23)		(-2.38)		(1.35)
SIZE		-0.475		-0.244		-0.355		-0.269		-0.266		-0.328
		(-2.61)		(-2.08)		(-3.38)		(-2.67)		(-2.63)		(-3.22)
BM		-0.101		0.234		0.257		0.385		0.268		0.345
		(-0.39)		(1.27)		(1.55)		(2.55)		(1.65)		(2.28)
Continued on next page												
Continued												
MOM		0.515		-0.084		0.436		0.326		0.608		0.336
		(1.68)		(-0.19)		(1.86)		(0.98)		(2.89)		(1.28)
REV		-0.032		-0.013		-0.012		-0.011		-0.012		-0.007
		(-2.51)		(-1.03)		(-1.10)		(-1.32)		(-1.29)		(-0.90)
ILLIQ		-0.186		-0.090		-0.146		0.028		0.050		0.016
		(-1.04)		(-0.46)		(-1.37)		(0.46)		(0.68)		(0.21)
Variables	DSKC% Split											
	40th				50th				Mean			
	Sunny		Cloudy		Sunny		Cloudy		Sunny		Cloudy	
MAX	-0.043	-0.049	-0.007	-0.008	-0.047	-0.043	-0.025	-0.019	-0.049	-0.052	-0.026	-0.020
	(-3.17)	(-2.78)	(-0.51)	(-0.47)	(-3.21)	(-3.17)	(-1.95)	(-1.19)	(-3.42)	(-3.78)	(-1.95)	(-1.33)
BETA		-0.097		-0.062		-0.105		-0.047		-0.085		-0.072
		(-1.65)		(-1.18)		(-2.04)		(-1.00)		(-1.65)		(-1.44)
SIZE		-0.305		-0.181		-0.269		-0.177		-0.318		-0.167
		(-3.97)		(-2.40)		(-3.67)		(-2.60)		(-4.22)		(-2.43)
BM		0.257		0.367		0.233		0.286		0.238		0.240
		(1.98)		(3.10)		(1.91)		(2.75)		(1.93)		(2.38)
MOM		0.468		0.452		0.506		0.453		0.490		0.438
		(2.03)		(2.46)		(2.30)		(2.38)		(2.06)		(2.32)
REV		-0.023		-0.007		-0.016		-0.010		-0.016		-0.009
		(-3.52)		(-1.19)		(-2.68)		(-1.67)		(-2.85)		(-1.49)
ILLIQ		0.094		0.022		0.127		0.029		0.133		0.021
		(1.78)		(1.26)		(1.78)		(1.29)		(1.75)		(0.91)

DSKC, respectively. The multivariate results following cloudy weather conditions suggest that the MAX effect is statistically insignificant at conventional levels. The exception is when we use the 25th percentile of DSKC as the threshold. The coefficient for MAX is −0.042 and significant (*t*-statistic = −1.96); however, it is less significant both economically and statistically than the corresponding coefficient (−0.050, *t*-statistic = −2.48) for sunny weather conditions.

In summary, Table 5 shows that the MAX effect is stronger economically and statistically following pleasant weather conditions (sunny months with low DSKC) than unpleasant weather conditions (cloudy months with high DSKC), indicating that lottery demand is influenced by weather conditions. The results provide further support for the existing psychological literature that documents asymmetric effects of good (optimistic) and bad (pessimistic) moods, where biases caused by good mood are stronger than those caused by bad mood (Chhaochharia et al., 2019; Schwarz, 1990). A good mood can lead to reliance on non-systematic cognitive systems, which induce individuals to pay less attention to details. However, a bad mood can induce individuals to pay more attention to the details of the task; consequently, people tend to be less affected by mood heuristics (Schwarz, 1990; Petty, Gleicher, and Baker, 1991; Sinclair and Mark, 1995).

4.3. Cross-sectional regressions with an interaction term

In the previous section, we divide the sample periods into subsamples of sunny and cloudy weather months. It is important to show that our results are robust when we run the cross-sectional regression across the whole sample. In this section, we re-run the Fama and MacBeth (1973) cross-sectional regressions after controlling for DSKC and an interaction term between MAX and a corresponding dummy variable that differentiates between firms with low and high values on DSKC, without and with control variables. This approach allows us to study how the MAX effect varies between firms under different weather conditions.

We run the monthly cross-sectional regression Eq. (2):

$$R_{i,t} = \lambda_{0,t-1} + \lambda_{1,t-1}MAX_{i,t-1} + \lambda_{2,t-1}DSKC_{i,t-1} + \lambda_{3,t-1}MAX_{i,t-1} \times DDSKC_{i,t-1} + \boldsymbol{\Lambda}_{t-1}\boldsymbol{Z}_{i,t-1} + \mathscr{E}_{i,t} \quad (2)$$

where $R_{i,t}$ is the realized return (in percent) on stock i in month t, $MAX_{i,t-1}$ is the maximum daily return for stock i in month $t-1$ and $DSKC_{i,t-1}$ is the deseasonalized sky cloud cover measure for the MSA where firm i (i.e., stock i's firm) is headquartered in month $t-1$. $DDSKC_{i,t-1}$ is a dummy variable defined based on DSKC of the MSA where firm i is headquartered. If the firm's associated DSKC in month $t-1$ is below the thresholds of DSKC (sunny weather), the $DDSKC_{i,t-1}$ is

one and zero otherwise. The thresholds are the mean and median of DSKC of the firm's headquarter metropolitan city over time. $\boldsymbol{Z}_{i,t-1}$ is a vector of control variables that include market beta (BETA), natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). These control variables are explained in the Appendix. Table 6 reports the time-series averages of the slope coefficients. Their corresponding *t*-statistics are adjusted using the Newey-West (1987) method with six lags.

Panel A of Table 6 shows the results where the median of DSKC is the threshold used to define DDSKC. The average slope coefficient of MAX for the univariate regression (model (1)) is −0.028 (*t*-statistic = −2.87). This coefficient becomes stronger both economically and statically after taking into account the control variables in model (2) (−0.030, *t*-statistic = −3.26). These results are consistent with previous studies that document the MAX effect in the US (e.g., Bali et al., 2011, among others), indicating that there is a statistically significant and negative relationship between the maximum daily return over the previous month and the cross-sectional expected stock returns.

In model (3), we include only DSKC and MAX. The average slope coefficient on DSKC is negative and insignificant, whereas the slope coefficient on MAX remains very close to those in models (1) and (2) and statistically significant (−0.029, *t*-statistic = −2.95). This finding is robust when we incorporate control variables in model (4). In both models, the negative sign of the slope coefficient on DSKC is consistent with the weather effect documented in the literature, suggesting that there is a negative relationship between weather-induced pessimistic mood (cloudy weather) and stock returns, or equivalently, a positive relation between weather-induced optimistic mood (sunny weather) and stock returns (e.g., Saunders, 1993; Hirshleifer and Shumway, 2003).

Models (5) to (8) report the main results. Models (5) and (6) include MAX, DSKC and the interaction term between MAX and the dummy variable for DSKC (DDSKC) before and after controls. Since our primary research question revolves around the interaction effect between MAX

Table 6

Fama-MacBeth regressions with an interaction term. Every month from January 1983 to December 2019, we run the Fama-MacBeth regression of the monthly stock returns (in percent) on the one-month lagged independent variables that include the maximum daily return (MAX), DSKC, the interaction term of MAX and DDSKC, and control variables. DDSKC is a dummy variable based on DSKC within each MSA where firms are headquartered. If a firm's monthly DSKC is below the threshold of DSKC for its corresponding MSA over time, then DDSKC is one and zero otherwise. The control variables are market beta (BETA), natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). All variables are defined in the Appendix. The table reports the time-series average of the slope coefficients from cross-sectional regressions across the sample when the threshold to define DDSKC is median (Panel A) and mean (Panel B) of DSKC within each MSA over time. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

Panel A: Median of DSKC as the threshold to define DDSKC								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MAX	-0.028	-0.030	-0.029	-0.029	-0.023	-0.015		
	(-2.87)	(-3.26)	(-2.95)	(-3.12)	(-1.99)	(-1.10)		
DSKC			-0.005	-0.018	-0.148	-0.125		
			(-0.05)	(-0.23)	(-1.45)	(-1.30)		
MAX × DDSKC					-0.022	-0.027	-0.028	-0.018
					(-1.97)	(-1.74)	(-2.65)	(-1.82)
BETA		-0.041		-0.051		-0.051		-0.065
		(-1.18)		(-1.46)		(-1.48)		(-1.79)
SIZE		-0.236		-0.237		-0.242		-0.213
		(-3.97)		(-3.83)		(-3.91)		(-3.49)
Continued on next page…								
…Continued								
BM		0.204		0.229		0.217		0.227
		(2.21)		(2.46)		(2.33)		(2.30)
MOM		0.375		0.374		0.377		0.400
		(2.35)		(2.36)		(2.38)		(2.46)
REV		-0.012		-0.012		-0.012		-0.011
		(-2.84)		(-2.88)		(-2.91)		(-2.72)
ILLIQ		0.021		0.024		0.024		0.023
		(3.38)		(3.38)		(3.36)		(3.32)
Panel B: Mean of DSKC as the threshold to define DDSKC								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MAX	-0.028	-0.030	-0.029	-0.029	-0.024	-0.019		
	(-2.87)	(-3.26)	(-2.95)	(-3.12)	(-2.07)	(-1.48)		
DSKC			-0.005	-0.018	-0.156	-0.137		
			(-0.05)	(-0.23)	(-1.45)	(-1.40)		
MAX × DDSKC					-0.022	-0.023	-0.029	-0.019
					(-1.92)	(-1.56)	(-3.08)	(-1.99)
BETA		-0.041		-0.051		-0.051		-0.067
		(-1.18)		(-1.46)		(-1.47)		(-1.82)
SIZE		-0.236		-0.237		-0.241		-0.210
		(-3.97)		(-3.83)		(-3.90)		(-3.43)
BM		0.204		0.229		0.219		0.231
		(2.21)		(2.46)		(2.35)		(2.34)
MOM		0.375		0.374		0.378		0.398
		(2.35)		(2.36)		(2.41)		(2.44)
Continued on next page…								
…Continued								
REV		-0.012		-0.012		-0.012		-0.011
		(-2.84)		(-2.88)		(-2.91)		(-2.69)
ILLIQ		0.021		0.024		0.024		0.023
		(3.38)		(3.38)		(3.35)		(3.32)

and weather, and we are specifically interested in examining whether the MAX effect varies across different levels of the DDSKC, we also provide results that only include the interaction term without and with control variables in models (7) and (8), respectively.

The slope coefficient for the interaction term between MAX and DDSKC indicates the relation between the MAX effect and the weather-induced mood, implying the economic difference of the MAX effect between firms with low (sunny) and high values (cloudy) on DSKC. The coefficients are negative and statistically significant across all specifications, suggesting that sunny weather contributes to the MAX effect. For instance, in model (5), the coefficient estimate on MAX is −0.023 (*t*-statistic = −1.99) and the interaction term ($MAX \times DDSKC$) coefficient estimate is −0.022 (*t*-statistic= −1.97), which are both significant at the 5% level. Since the DDSKC in the interaction term equals to one for firms with low values on DSKC (sunny) and equals to zero for firms with high values on DSKC (cloudy), the results imply that the MAX effect is weaker (coefficient estimate = −0.023), albeit statistically significant (*t*-statistic = −1.99), among firms with high DSKC (following cloudy weather), whereas the MAX effect tends to become stronger among firms with low DSKC (following sunny weather), with the estimated coefficient of −0.045 (formally, −0.023 + (−0.022) × 1). Model (6) is the full model that includes all control variables. Similarly, the estimated coefficient of the interaction term ($MAX \times DDSKC$) is −0.027 (*t*-statistic = −1.74), which is significant at the 10% level. Consistent with the results shown in model (5), the MAX effect is weaker and statistically insignificant (coefficient estimate = −0.015 and *t*-statistic = −1.10) among firms with high DSKC. Nonetheless, the MAX effect is stronger among firms with low DSKC, with the estimated coefficient of −0.042 (formally, −0.015 + (−0.027) × 1). These results indicate that the MAX effect is doubled following sunny weather.

Since our primary research question is around the interaction effect between MAX and weather, we run Eq. (2) with only the interaction term in models (7) and (8). The estimated coefficients of the interaction term in both specifications are significantly negative, with values of −0.028 (*t*-statistic = −2.65) and −0.018 (*t*-statistic = −1.82), respectively. The results support our finding that MAX effect relates to weather conditions and that it is stronger following sunny weather.

We also use the mean of the DSKC as the threshold to define DDSKC, and repeat the analysis done in panel A. The results, reported in panel B, are qualitatively similar to those in Panel A. Overall, the results reported in Table 6 provide further support for our hypothesis that weather plays an important role in MAX effect and lottery demand. In particular, the significantly negative coefficients on the interaction term ($MAX \times DDSKC$) provide evidence that lottery-like stocks with low DSKC (sunny) have lower returns compared with lottery-like stocks with high DSKC (cloudy), indicating that the MAX effect is stronger following pleasant weather conditions (sunny weather with low DSKC).

4.4. Rain, wind and temperature as alternative weather-induced mood proxies

So far, we show the significant role of weather-induced mood, proxied by the sunshine variable (DSKC), in influencing the MAX effect. Similar to cloudy weather, rainy and windy conditions are considered unpleasant and can lead to negative mood states, which involve negative effects such as stronger pessimism, reduced activity and underreaction to information events (e.g., Dehaan et al., 2017; Jiang et al., 2021). Moreover, psychological research has shown that temperature influences mood and behavior positively (e.g., Cunnigham 1979; Howarth and Hoffman, 1984; Keller et al., 2005, Hu and Lee, 2020).

In this section, as robustness tests, we investigate if rain, wind and temperature as alternative proxies for weather-induced mood can impact the MAX effect. We define RAIN, WIND, and TEMP as the liquid precipitation depth dimension, wind speed rate, and air temperature, respectively. Similar to DSKC, these weather variables are measured for the MSAs where the firms are headquartered. The detailed information of these variables is provided in the Appendix. Our approach using the rain and wind variables is similar, and explained first. We discuss the method and results for temperature in a separate subsection afterwards.

4.4.1. Rain and wind as the weather-induced mood proxies

We repeat the analysis in Table 6 using RAIN and WIND variables instead of DSKC.[22] To do so, firstly, we define dummy variables DRain and DWind based on RAIN and WIND variables of the firm's headquarter metropolitan city, respectively. If the firm's associated RAIN (WIND) in a month is below the mean of RAIN (WIND) of the headquarter metropolitan city over time, the DRain (DWind) is one and zero otherwise.[23] The value of one for DRain (DWind) indicates the weather is pleasant, whereas the value of zero suggests it is unpleasant. Then, we run the cross-sectional regression Eq. (2) using one-month lagged RAIN and DRain (WIND and DWind) instead of DSKC and DDSKC, respectively. Table 7 presents the results.

Panel A and Panel B show the results for RAIN and WIND used as the weather-induced mood proxy, respectively. In models (1) and (2) of Panel A in Table 7, the slope coefficients on MAX remain significant in the presence of the RAIN variable. When we introduce the interaction term in models (3) and (4), the estimated coefficients on MAX are reduced significantly, both economically and statistically, but those of the interaction terms are relatively large. For example, in model (4) the coefficient estimate on MAX is −0.008 (*t*-statistic = −0.52); however, the coefficient for the interaction term ($MAX \times DRain$) is −0.028 (*t*-statistic = −1.62), which is very close to the slope coefficient of MAX in model (2). The results indicate that the MAX effect is weaker (coefficient estimate = −0.008) and statistically insignificant (*t*-statistic = −0.52) among firms following unpleasant weather conditions (DRain = 0). However, it becomes stronger among firms following pleasant weather conditions (DRain = 1), with the estimated coefficient of −0.036 (formally, −0.008 + (−0.028) × 1). Due to multicollinearity concerns between MAX and the interaction term, we also report the results for the specifications that include only the interaction term without and with control variables in models (5) and (6). The slope coefficients on the interaction term ($MAX \times DRain$) are −0.022 (*t*-statistic = −2.40) and −0.028 (*t*-statistic = −2.53), respectively, suggesting that the MAX effect is stronger following pleasant weather.

The results in Panel B of Table 7 show qualitatively similar results when we use wind as the proxy for weather-induced mood. The MAX effect is significant in presence of the WIND variable, but becomes less significant, both economically and statistically, when we include the interaction term ($MAX \times DWind$) in the models. The significant slope coefficients of the interaction term indicate that there is a relation between WIND and the MAX effect. For example, the results in model (4) of Panel B show that the MAX effect is weaker (coefficient estimate = −0.020) among firms following unpleasant weather conditions (DWind = 0). However, it is stronger among firms following pleasant weather conditions (DWind = 1), with an estimated coefficient of −0.038 (formally, −0.020 + (−0.018) × 1).

These findings are robust when we only include the interaction term between MAX and DWind without and with control variables in models (5) and (6) in Panel B. The estimated coefficients of the interaction term ($MAX \times DWind$) are −0.028 (*t*-statistic = −3.17) and −0.021 (*t*-statistic = −2.33), respectively, suggesting that pleasant weather conditions, proxied by the WIND variable, can drive the MAX effect.

The results in Table 7 provide further support to our findings in

[22] In robustness tests, we show that our findings are robust when we use deseasonalized RAIN and WIND in the analyses in Table IA11 of the Internet Appendix.

[23] In unreported analyses, we also use median of RAIN and WIND as the threshold to define the dummy variables. The results are robust, though they are weaker for RAIN because the number of months with zero rain biases the median values towards very low values in some MSAs.

Table 7
Weather-induced mood and MAX effect: rain and wind as the weather-induced mood proxies. Every month from January 1983 to December 2019, we run the Fama-MacBeth regression of the monthly stock returns (in percent) on the one-month lagged independent variables that include the maximum daily return (MAX), RAIN (WIND), the interaction term of MAX and DRain (DWind), and control variables. DRain (DWind) is a dummy variable based on RAIN (WIND) within each MSA where firms are headquartered. If a firm's monthly value of RAIN (WIND) is below the mean of RAIN (WIND) for its corresponding MSA over time, then DRain (DWind) is one and zero otherwise. The control variables are market beta (BETA), natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). All variables are defined in the Appendix. Panel A (Panel B) reports the time-series average of the slope coefficients from cross-sectional regressions across the sample when RAIN (WIND) is used as the proxy for weather-induced mood. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

	Panel A: RAIN as the weather-induce mood proxy						Panel B: WIND as the weather-induced mood proxy					
Model	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
MAX	-0.027	-0.032	-0.016	-0.008			-0.028	-0.030	-0.018	-0.020		
	(-2.69)	(-3.35)	(-1.19)	(-0.52)			(-2.87)	(-3.28)	(-1.55)	(-1.78)		
RAIN	0.059	0.070	-0.041	-0.098								
	(0.80)	(0.93)	(-0.45)	(-1.03)								
MAX × DRain			-0.012	-0.028	-0.022	-0.028						
			(-1.05)	(-1.62)	(-2.40)	(-2.53)						
WIND							0.001	-0.007	-0.001	-0.007		
							(0.23)	(-1.77)	(-0.15)	(-1.59)		
MAX × DWind									-0.017	-0.018	-0.028	-0.021
									(-1.67)	(-1.68)	(-3.17)	(-2.33)
Continued on next page												
Continued												
BETA		-0.046		-0.048		-0.047		-0.044		-0.042		-0.046
		(-1.34)		(-1.37)		(-1.31)		(-1.25)		(-1.21)		(-1.29)
SIZE		-0.280		-0.277		-0.267		-0.243		-0.247		-0.232
		(-4.29)		(-4.26)		(-4.05)		(-4.01)		(-4.08)		(-3.86)
BM		0.227		0.230		0.224		0.215		0.219		0.226
		(2.26)		(2.30)		(2.12)		(2.30)		(2.37)		(2.30)
MOM		0.406		0.414		0.412		0.386		0.378		0.404
		(2.59)		(2.66)		(2.57)		(2.43)		(2.37)		(2.50)
REV		-0.013		-0.013		-0.013		-0.012		-0.012		-0.011
		(-3.21)		(-3.27)		(-3.09)		(-2.89)		(-2.93)		(-2.78)
ILLIQ		0.020		0.020		0.019		0.021		0.021		0.020
		(3.02)		(3.09)		(3.14)		(3.33)		(3.31)		(3.23)

Table 6 that the MAX effect mainly exists following relatively pleasant weather conditions (less rain or less wind) rather than relatively unpleasant weather conditions (more rain or more wind). The results also support the existing literature that unpleasant weather such as cloudy, rainy and windy conditions can negatively affect human moods and behavior, consistent with the prior findings of the impacts of unpleasant weather mentioned by Dehaan et al. (2017) as well as Jiang et al. (2021).

4.4.2. *Temperature as the weather-induced mood proxy*

Now we turn to investigate the relation between the MAX effect and the weather-induced mood measured by temperature. Psychological evidence suggests that higher temperature triggers a good mood (e.g., Cunnigham 1979; Howarth and Hoffman, 1984). Consistent with this evidence, Keller et al. (2005) show that higher temperature induces a positive mood in investors and leads to risk-taking behavior. Also, Hu and Lee (2020) demonstrate that higher temperature, as a proxy for weather-induced positive mood, is associated with higher transaction prices in the housing market.

In this section, we explore whether higher levels of temperature impact the MAX effect. We postulate that higher temperature, as a pleasant weather, is related to a positive mood and can induce risk-taking behavior that leads to higher lottery demand and MAX effect.

The monthly maximum temperature for each headquarter metropolitan city of a firm, denoted TEMP, is calculated as the average of the daily maximum temperature levels over the month. Detailed definition of the variable is provided in the Appendix. We run the cross-sectional regression Eq. (2) using one-month lagged of TEMP instead of DSKC and DDSKC and repeat the analysis done in Table 6.[24] Table 8 presents the results.

Coefficient estimates of TEMP and its interaction term are multiplied by 100 for display purposes. Our main interests are the slope coefficients on MAX, and the interaction term between MAX and TEMP, which shows whether higher levels of temperature can impact the MAX effect.

In models (1) and (2), the slope coefficients on MAX remain significantly negative with values of −0.028 and −0.031 and corresponding *t*-statistics of −2.89 and −3.41, respectively, when we include TEMP in the specifications. In models (3) and (4), once we introduce the interaction term between MAX and TEMP, the estimated slope coefficients of MAX become insignificant. The coefficients of the interaction term are also insignificant, which might be due to the multicollinearity between MAX and the interaction term. Therefore, we run models (5) and (6), which include only the interaction term with and without control variables, respectively. In models (5) to (6), we find that the slope coefficients of the interaction term $(MAX \times TEMP)$ with and without control variables are significantly negative, with values of −0.010 (*t*-statistic = −1.65) and −0.012 (*t*-statistic = −1.95), respectively. This means that higher levels of temperature make the MAX effect stronger, which is consistent with the notion that higher temperature is associated with a positive mood (e.g., Cunnigham 1979; Howarth and Hoffman, 1984; Hu and Lee, 2020) and can lead to investors' risk-taking behavior (Keller et al., 2005).

Overall, the results reported in Table 8 provide additional support for our findings in Table 6. We find that higher temperature, as a pleasant weather condition, can play an important role in the MAX effect.

4.5. *Further robustness tests*

We further conduct a series of additional analyses to check the robustness of our main results. The analyses are designed to demonstrate the robustness of the ability of weather-induced mood to impact the lottery demand. In this section, we summarize these additional tests and

[24] In robustness tests, we show that our results are robust when we use deseasonalized TEMP in the analyses in Table IA12 of the Internet Appendix.

Table 8
Weather-induced mood and MAX effect: temperature as the weather-induced mood proxy. Every month from January 1983 to December 2019, we run the Fama-MacBeth regression of the monthly stock returns (in percent) on the one-month lagged independent variables that include the maximum daily return (MAX), TEMP, the interaction term of MAX and TEMP, and control variables. TEMP is the monthly maximum temperature for each MSA where the firms are headquartered. The control variables are market beta (BETA), natural log of market capitalization (SIZE), natural log of book-to-market ratio (BM), momentum (MOM), short-term reversal (REV), and illiquidity (ILLIQ). All variables are defined in the Appendix. The table reports the time-series average of the slope coefficients from cross-sectional regressions across the sample when TEMP is used as the proxy for weather-induced mood. The coefficient estimates of TEMP and its interaction term are multiplied by 100 for display purposes. The sample includes Nasdaq-listed stocks headquartered in the 10 largest US MSAs. *t*-statistics are adjusted using Newey and West (1987) standard errors with six lags and reported in parentheses.

TEMP as the weather-induced mood proxy						
Model	(1)	(2)	(3)	(4)	(5)	(6)
MAX	-0.028	-0.031	-0.030	-0.010		
	(-2.89)	(-3.41)	(-1.19)	(-0.33)		
TEMP	-0.500	-0.400	-0.400	-0.300		
	(-2.63)	(-2.28)	(-2.02)	(-1.40)		
MAX × TEMP			-0.003	-0.009	-0.010	-0.012
			(-0.25)	(-0.50)	(-1.65)	(-1.95)
BETA		-0.043		-0.044		-0.044
		(-1.25)		(-1.26)		(-1.23)
Continued on next page…						
…Continued						
SIZE		-0.250		-0.245		-0.243
		(-4.12)		(-4.05)		(-4.06)
BM		0.217		0.215		0.214
		(2.36)		(2.33)		(2.21)
MOM		0.371		0.365		0.379
		(2.33)		(2.30)		(2.34)
REV		-0.012		-0.012		-0.012
		(-2.93)		(-2.90)		(-2.90)
ILLIQ		0.021		0.022		0.021
		(3.31)		(3.32)		(3.32)

provide a detailed discussion in the Internet Appendix. The relevant sections and tables of the Internet Appendix are presented in parentheses. First, we show that the main results are robust when MAX is replaced with MAX(N), which is the average of N (N = 2, 3, 4, 5) highest daily returns over the past one month, as the alternative measures of lottery demand in the cross-sectional regressions (Tables IA1 and IA2 of Section IA-1). We also demonstrate that the ability of weather-induced mood to influence the lottery demand continue to hold when skewness measures are used as the proxies for lottery demand, instead of MAX, in the cross-sectional regressions (Tables IA3 and IA4 of Section IA-2). Then, we show that the MAX effect cannot be detected when performing a univariate portfolio analysis using the component of MAX that is orthogonal to the weather-induced mood proxy DSKC as the sort variable (Table IA5 of Section IA-3). In our main univariate portfolio analysis that examines the relation between weather-induced mood and the lottery demand, we define DSKC ranking (threshold) within each MSA over time. We show that our results are robust when using the ranking based on full sample (i.e., across all MSAs where firms are headquartered) in each month (Table IA6 of Section IA-4). Next, we demonstrate that our main results persist when replacing DSKC with SKC, which is not deseasonalized, as the weather-induced mood proxy in both portfolio analysis (Tables IA7 and IA8 of Section IA-5) and the cross-sectional regressions (Tables IA9 and IA10 of Section IA-5). Finally, we use deseasonalized measures of RAIN and WIND variables (Table IA11 of Section IA-6) and TEMP variable (Table IA12 of Section IA-6) as the alternative proxies for weather-induced mood in the cross-sectional regressions.

4.6. The potential economic channel

In previous sections, we demonstrated the impact of the weather-induced mood proxied by several weather variables on the MAX effect. In this section, we provide a discussion about a potential channel through which weather affects mood and consequently the lottery demand and the underperformance of lottery-type stocks.

We conjecture that risk-taking is the channel through which weather via mood affects the lottery demand. The idea is that investors with a weather-induced positive mood tend to take more risks and are attracted to lottery stocks, which are riskier than non-lottery stocks. Higher demand exerts upward pressure on the price of these stocks, which results in higher prices and lower expected returns for lottery stocks. Our argument is based on the prior research that suggests mood affects behavior through the risk preferences (or attitudes) channel. For example, Cortés et al. (2016) show that loan officers with a positive mood, induced by sunshine, tend to take more risks and approve more risky loan applications. This results in higher credit (or loan) approvals on sunny days rather than cloudy days. Moreover, Bassi et al. (2013) investigate the relation between weather, mood and risk-taking behavior in financial decisions, and find that sunshine and good weather can enhance financial risk-taking behavior through their effect on mood. Grable and Roszkowski (2008) also show that being in a happy mood is positively related with having a higher level of willingness to take financial risks.

To test our conjecture directly, we need to analyze buy and sell trade data and examine the trades quantity and direction (i.e., buy and sell orders, and buy-sell order imbalances) of lottery-type stocks.[25] However, due to lack of data we were not able to examine this conjecture, so we leave the related empirical test for the future research.

5. Conclusion

Extant research has shown that investors have a strong preference for lottery-like stocks (e.g., Kumar, 2009; Bali et al., 2011). With the high demand for lottery-like stocks, these stocks tend to be overvalued and exhibit low future returns. This underperformance of lottery-like stocks is consistent with the model suggested in Barberis and Huang (2008) based on the cumulative prospect theory (Tversky and Kahneman, 1992) and the optimal beliefs theory of Brunnermeier et al. (2007).

We conjecture that weather can influence the underperformance of lottery-like stocks through its effect on mood. Psychology literature documents that pleasant or unpleasant weather conditions, such as sunshine or lack of sunshine, affect mood (e.g., Persinger, 1975; Cunningham, 1979; Schwarz and Clore, 1983; Parrott and Sabini, 1990; Scott, 2007), and that people who are in good (bad) moods tend to have more optimistic (pessimistic) evaluations of their environment as their mood influences their judgements and choices (e.g., Isen et al., 1978; Schwarz and Clore, 1983; Johnson and Tversky, 1983; Wright and Bower, 1992; Forgas, 1995); they also exhibit greater (less) risk-taking (e.g., Bassi et al., 2013; Cortés et al., 2016), have higher (lower) demands for or intention to buy (sell) stocks (e.g., Loughran and Schultz, 2004; Goetzmann and Zhu, 2005; Goetzmann et al., 2015; Chang et al., 2008), and generate a positive (negative) influence on the stock market (e.g., Saunders, 1993; Hirshleifer and Shumway, 2003; Goetzmann and Zhu, 2005). Consistent with this psychology literature, we provide compelling evidence that supports our conjecture.

We use the maximum daily return (MAX) over the last month as the main proxy for lottery-type stock and deseasonalized sky cloud cover (DSKC) as the main proxy for weather-induced mood. We find that the MAX effect is stronger following sunny weather conditions compared to cloudy weather conditions in both portfolio analysis and the cross-

[25] We thank an anonymous referee for their insightful comment about the required analysis.

sectional tests. Moreover, we introduce rain, wind and temperature as the proxies for weather conditions in the cross-sectional tests. The results of rain wind, and temperature provide additional evidence on those related to DSKC, showing that MAX effect is more prevalent after relatively pleasant weather (less rain, less wind or higher temperature) than unpleasant weather (more rain, more wind or lower temperature) months. We further conduct a battery of robustness tests in the Internet Appendix and show that our main results are robust. Our paper provides insights into how mood affects investors' financial decisions and ultimately financial market outcomes.

CRediT authorship contribution statement

Reza Bradrania: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ya Gao:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Appendix: Variable Definitions

MAX: The maximum daily return within a month as in Bali et al. (2011), which can be calculated using the following equation:

$$\mathrm{MAX}_{\mathrm{i,t}} = \max\left(R_{i,d}\right), \mathrm{d} = \{1, 2, \ldots D_t\}, \tag{A1}$$

where $R_{i,d}$ is the return of stock i on day d and D_t is the number of trading days in month t.

MAX(N): the average of N (N = 2, 3, 4, 5) highest daily returns within a month, following Bali et al. (2011).

BETA: The market beta over the month. Following Scholes and Williams (1977) and Dimson (1979), we consider nonsynchronous trading and use the lag, current and lead market return when estimating beta. We estimate beta using daily returns within each month for each stock:

$$\mathrm{R}_{\mathrm{i,d}} - \mathrm{r}_{\mathrm{f,d}} = \alpha_{\mathrm{i}} + \beta_{1,\mathrm{i}}\left(R_{m,d-1} - r_{f,d-1}\right) + \beta_{2,\mathrm{i}}\left(R_{m,d} - r_{f,d}\right) + \beta_{3,\mathrm{i}}\left(R_{m,d+1} - r_{f,d+1}\right) + \varepsilon_{i,d}, \tag{A2}$$

where $R_{i,d}$ is the return of stock i on day d and $R_{m,d}$ is the market return on day d, which is measured by the CRSP daily value-weighted index, and $r_{f,d}$ is the risk-free rate on day d, which is the one-month T-bill return obtained from Kenneth French's online data library. The market beta of stock i in month t is computed as $\widehat{\beta_{i,t}} = \widehat{\beta_{1,i}} + \widehat{\beta_{2,i}} + \widehat{\beta_{3,i}}$.

MKTCAP: The market value of equity, which is measured by a stock's price multiplied by shares outstanding in millions of dollars at the end of month $t-1$ for each stock.

SIZE: The natural log of market capitalization (MKTCAP).

BTM: Following Fama and French (1992), a firm's book-to-market ratio in month t is calculated by using the market value of equity at the end of December of the previous year and the book value at the end of the previous fiscal year. The book value of equity is calculated from the balance sheet data from Compustat, which is defined as stockholder's equity adding deferred taxes and investment tax credit then subtracting the book value of preferred stock. The book value of preferred stock is taken to be the redemption value, the liquidating value or the par value, taken as available in the given order.

BM: The natural log of book-to-market ratio (BTM).

MOM: Following Jegadeesh and Titman (1993), momentum for each stock in month t is defined as the cumulative return on the stock over the previous 11 months: here, starting two months ago, namely, the cumulative return from month $t-12$ to month $t-2$, inclusive.

REV: The short-term reversal for each stock in month t, which is defined as the return on the stock over the past one month based on Jegadeesh (1990); that is, the return in month $t-1$.

ILLIQ: Following Amihud (2002), the illiquidity for each stock in month t is measured by using the monthly average of the absolute daily stock return divided by its dollar trading volume:

$$\mathrm{ILLIQ}_{\mathrm{i,t}} = \frac{1}{D}\sum_{d=1}^{D}\frac{|R_{i,d}|}{VOLD_{i,d}}, \tag{A3}$$

where $R_{i,d}$ is the return of stock i on day d and $VOLD_{i,d}$ is the daily trading volume in dollars for each stock, measured by volume times the stock price in millions of dollars. Following Lin and Liu (2018), we require the trading volume data to be available for a minimum of ten days each month to compute the illiquidity.

TSKEW: The total skewness of stock i for month t, as a scaled measure of the third moment of daily returns, is computed by using daily returns within the month t:

$$\mathrm{TSKEW}_{\mathrm{i,t}} = \frac{1}{D_t}\sum_{d=1}^{D_t}\left(\frac{R_{i,d} - \mu_i}{\sigma_i}\right)^3, \tag{A4}$$

where D_t is the number of trading days in month t, $R_{i,d}$ is the return of stock i on day d, μ_i is the mean of returns for stock i in month t, and σ_i is the standard deviation of daily returns for stock i in month t.

ISKEW: Following Bali et al. (2011), the idiosyncratic skewness is estimated from the following regression for each stock:

$$\mathrm{R}_{\mathrm{i,d}} - \mathrm{r}_{\mathrm{f,d}} = \alpha_{\mathrm{i}} + \beta_{\mathrm{i}}\left(R_{m,d} - r_{f,d}\right) + \gamma_{\mathrm{i}}\left(R_{m,d} - r_{f,d}\right)^2 + \varepsilon_{i,d}, \tag{A5}$$

where $R_{i,d}$ is the return of stock i on day d, $R_{m,d}$ is the market return on day d, which is measured by the CRSP daily value-weighted index, $r_{f,d}$ is the risk-free rate on day d, which is the one-month T-bill return obtained from Kenneth French's online data library, and $\varepsilon_{i,d}$ is the idiosyncratic return on

day d. The idiosyncratic skewness (ISKEW) of stock i in month t is computed as the skewness (or third moment) of daily residuals $\varepsilon_{i,d}$ in month t.

Two main weather-induced mood proxies related to sunshine:

SKC: Sky cloud cover, the hourly SKC measures the total celestial dome covered by clouds, ranging from 0 (clear or the least cloudy weather) to 8 (overcast or the cloudiest weather). For each MSA, we retain SKC observations between the hours of 6 am to 4 pm and require the data to be available for a minimum of three hours each day and a minimum of 11 days each month. The daily SKC for each MSA is computed as the average of hourly SKC values of each day. Since MAX is at a monthly frequency, we calculate monthly SKC for each MSA by averaging the daily estimates over the month.

DUMSKC: A dummy variable based on SKC within each MSA where firms are headquartered. If a firm's monthly SKC value is below (sunny) the thresholds (median and mean) of SKC of its corresponding MSA over time, then DUMSKC is equal to one and zero otherwise.

DSKC: Deseasonalized sky cloud cover, which is widely used in weather effect study to alleviate the seasonal weather patterns. Following Goetzmann and Zhu (2005), DSKC of month t for each MSA is defined as the SKC in month t minus the average SKC over the same month t across the entire sample period. In our study, it is a 37-year sample period.

DDSKC: A dummy variable based on DSKC within each MSA where firms are headquartered. If a firm's monthly DSKC value is below (sunny) the thresholds (10th, 20th, 25th, 40th, 50th percentiles and mean) of DSKC of its corresponding MSA over time, then DDSKC is equal to one and zero otherwise.

***MAX⊥DSKC*:** the portion of MAX of stock i in month t($\mathrm{MAX}_{i,t}$)that is orthogonal to (or that cannot be explained by) the weather variable in month t ($DSKC_{i,t}$), which can be estimated from the following regression of MAX on DSKC for each stock i over the time:

$$\mathrm{MAX}_{i,t} = \alpha_i + \beta_i DSKC_{i,t} + \varepsilon_{i,t}, \tag{A6}$$

where $\mathrm{MAX}_{i,t}$ is the maximum daily return of stock i in month t, $DSKC_{i,t}$ is the deseasonalized sky cloud cover for the MSA where firm of stock i is headquartered in month t, and $\varepsilon_{i,t}$ is the $MAX\perp DKSC$ of stock i in month t.

Three other weather-induced mood proxies:

RAIN: The depth of liquid precipitation, which is measured over a one-hour accumulation period and scaled by 10 with the units of millimeters. Like the SKC calculation, monthly RAIN for each MSA is calculated by averaging the daily estimates over the month after retaining observations between the hours of 6 am to 4 pm. The daily RAIN for each MSA is computed as the average of hourly RAIN values for each day. We require that non-missing data are available for a minimum of three hours each day and a minimum of 11 days each month when computing the monthly RAIN.

DRain: A dummy variable based on RAIN variable within each MSA where firms are headquartered. If a firm's monthly RAIN value is below (pleasant weather) the mean of RAIN value of its corresponding MSA over time, then DRain is equal to one and zero otherwise.

WIND: The hourly wind speed rate, which measures the rate of horizontal travel of air past a fixed point, is scaled by 10 with the units of meters per second. Similar to the calculation of SKC, monthly WIND for each MSA is calculated by averaging the daily estimates over the month after retaining observations between the hours of 6 am to 4 pm. The daily WIND for each MSA is computed as the average of hourly WIND values for each day. We require the wind data to be available for a minimum of three hours each day and a minimum of 11 days each month to compute the monthly WIND.

DWind: A dummy variable based on WIND variable within each MSA where firms are headquartered. If a firm's monthly WIND value is below (pleasant weather) the mean of WIND value of its corresponding MSA over time, then DWind is equal to one and zero otherwise.

TEMP: The maximum temperature of the air identified based on hourly temperature during the day and scaled by 10 with the units of degrees Celsius. Following the calculation of SKC, monthly TEMP for each MSA is the average of the daily maximum temperatures over the month after retaining the observations between the hours of 6 am to 4 pm. The daily TEMP for each MSA is the value of maximum hourly temperature for each day. We require that the observations are available for a minimum of three hours each day and a minimum of 11 days each month to compute the monthly TEMP.

Authors Statement

We wish to confirm that there are no known conflicts of interest associated with this paper and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We understand that the Corresponding Author is the sole contact for the Editorial process (including Editorial Manager and direct communications with the office). He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jbef.2024.100910.

References

Aboulamer, A., Kryzanowski, L., 2016. Are idiosyncratic volatility and MAX priced in the Canadian market? J. Empir. Financ. 37, 20–36.
Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. J. Financ. Mark. 5, 31–56.
An, L., Wang, H., Wang, J., Yu, J., 2020. Lottery-related anomalies: the role of reference-dependent preferences. Manag. Sci. 66, 473–501.
Anand, A., Gatchev, V.A., Madureira, L., Pirinsky, C.A., Underwood, S., 2011. Geographic proximity and price discovery: evidence from NASDAQ. J. Financ. Mark. 14, 193–226.
Annaert, J., De Ceuster, M., Verstegen, K., 2013. Are extreme returns priced in the stock market? European evidence. J. Bank. Financ. 37, 3401–3411.
Baker, M., Wurgler, J., 2006. Investor sentiment and the cross-section of stock returns. J. Financ. 61, 1645–1680.
Baker, M., Wurgler, J., 2007. Investor sentiment in the stock market. J. Econ. Perspect. 21, 129–151.

Bali, T.G., Brown, S.J., Murray, S., Tang, Y., 2017. A lottery-demand-based explanation of the beta anomaly. J. Financ. Quant. Anal. 52, 2369–2397.
Bali, T.G., Cakici, N., Whitelaw, R.F., 2011. Maxing out: stocks as lotteries and the cross-section of expected returns. J. Financ. Econ. 99, 427–446.
Bali, T.G., Hirshleifer, D., Peng, L., Tang, Y., 2019. Attention, social interaction, and investor attraction to lottery stocks. Unpubl. Work. Pap. Georget. Univ.
Barberis, N., Huang, M., 2008. Stocks as lotteries: the implications of probability weighting for security prices. Am. Econ. Rev. 98, 2066–2100.
Bassi, A., Colacito, R., Fulghieri, P., 2013. O sole mio: an experimental analysis of weather and risk attitudes in financial decisions. Rev. Financ. Stud. 26, 1824–1852.
Berggrun, L., Cardona, E., Lizarzaburu, E., 2019. Extreme daily returns and the cross-section of expected returns: evidence from Brazil. J. Bus. Res. 102, 201–211.
Bradrania, R., Veron, J.F., 2023. The beta anomaly in the Australian market and the lottery demand. Pac. -Basin Financ. J. 77, 101903.
Brunnermeier, M.K., Gollier, C., Parker, J.A., 2007. Optimal beliefs, asset prices and the preference for skewed returns. Am. Econ. Rev. 97, 159–165.
Brunnermeier, M.K., Parker, J.A., 2005. Optimal expectations. Am. Econ. Rev. 95, 1092–1118.
Byun, S.-J., Goh, J., Kim, D.-H., 2020. The role of psychological barriers in lottery-related anomalies. J. Bank. Financ. 114, 105786.
Campbell, J.Y., Hilscher, J., Szilagyi, J., 2008. In search of distress risk. J. Financ. 63, 2899–2939.
Carhart, M.M., 1997. On persistence in mutual fund performance. J. Financ. 52, 57–82.
Chang, S.-C., Chen, S.-S., Chou, R.K., Lin, Y.-H., 2008. Weather and intraday patterns in stock returns and trading activity. J. Bank. Financ. 32, 1754–1766.
Cheon, Y.-H., Lee, K.-H., 2018. Maxing out globally: individualism, investor attention, and the cross section of expected stock returns. Manag. Sci. 64, 5807–5831.
Chhaochharia, V., Kim, D., Korniotis, G.M., Kumar, A., 2019. Mood, firm behavior, and aggregate economic outcomes. J. Financ. Econ. 132, 427–450.
Cortés, K., Duchin, R., Sosyura, D., 2016. Clouded judgment: the role of sentiment in credit origination. J. Financ. Econ. 121, 392–413.
Coval, J.D., Moskowitz, T.J., 1999. Home bias at home: local equity preference in domestic portfolios. J. Financ. 54, 2045–2073.
Coval, J.D., Moskowitz, T.J., 2001. The geography of investment: informed trading and asset prices. J. Political Econ. 109, 811–841.
Cunningham, M.R., 1979. Weather, mood, and helping behavior: quasi experiments with the sunshine Samaritan. J. Personal. Soc. Psychol. 37, 1947–1956.
Dehaan, E., Madsen, J., Piotroski, J.D., 2017. Do weather-induced moods affect the processing of earnings news? J. Account. Res. 55, 509–550.
Dimson, E., 1979. Risk measurement when shares are subject to infrequent trading. J. Financ. Econ. 7, 197–226.
Fama, E.F., French, K.R., 1992. The cross-section of expected stock returns. J. Financ. 47, 427–465.
Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. J. Financ. Econ. 33, 3–56.
Fama, E.F., MacBeth, J.D., 1973. Risk and return: some empirical tests. J. Political Econ. 81, 607–636.
Fong, W.M., Toh, B., 2014. Investor sentiment and the MAX effect. J. Bank. Financ. 46, 190–201.
Forgas, J.P., 1995. Mood and judgment: the affect infusion model (AIM). Psychol. Bull. 117, 39–66.
Gao, Y., Bradrania, R., 2024. Property crime and lottery-related anomalies. Glob. Financ. J. 59, 1–24.
Goetzmann, W.N., Kim, D., Kumar, A., Wang, Q., 2015. Weather-induced mood, institutional investors, and stock returns. Rev. Financ. Stud. 28, 73–111.
Goetzmann, W.N., Zhu, N., 2005. Rain or shine: where is the weather effect? Eur. Financ. Manag. 11, 559–578.
Grable, J.E., Roszkowski, M.J., 2008. The influence of mood on the willingness to take financial risks. J. Risk Res. 11, 905–923.
Grinblatt, M., Keloharju, M., 2001. How distance, language and culture influence stockholdings and trades. J. Financ. 56, 1053–1073.
Hirshleifer, D., Shumway, T., 2003. Good day sunshine: stock returns and the weather. J. Financ. 58, 1009–1032.
Howarth, E., Hoffman, M.S., 1984. A multidimensional approach to the relationship between mood and weather. Br. J. Psychol. 75, 15–23.
Hu, M.R., Lee, A.D., 2020. Outshine to outbid: weather-induced sentiment and the housing market. Manag. Sci. 66, 1440–1472.
Huberman, G., 2001. Familiarity breeds investment. Rev. Financ. Stud. 14, 659–680.
Isen, A.M., Shalker, T.E., Clark, M., Karp, L., 1978. Affect, accessibility of material in memory and behavior: a cognitive loop? J. Personal. Soc. Psychol. 36, 1–12.
Ivković, Z., Weisbenner, S., 2005. Local does as local is: information content of the geography of individual investors' common stock investments. J. Financ. 60, 267–306.
Jegadeesh, N., 1990. Evidence of predictable behavior of security returns. J. Financ. 45, 881–898.
Jegadeesh, N., Titman, S., 1993. Returns to buying winners and selling losers: implications for stock market efficiency. J. Financ. 48, 65–91.
Jiang, D., Norris, D., Sun, L., 2021. Weather, institutional investors and earnings news. J. Corp. Financ. 69, 101990.
Johnson, E.J., Tversky, A., 1983. Affect, generalization, and the perception of risk. J. Personal. Soc. Psychol. 45, 20–31.
Keller, M.C., Fredrickson, B.L., Ybarra, O., Côté, S., Johnson, K., Mikels, J., Conway, A., Wager, T., 2005. A warm heart and a clear head. The contingent effects of weather on mood and cognition. Psychol. Sci. 16, 724–731.
Kumar, A., 2009. Who gambles in the stock market? J. Financ. 64, 1889–1933.
Lin, T.-C., Liu, X., 2018. Skewness, individual investor preference, and the cross-section of stock returns. Rev. Financ. 22, 1841-1876.
Loughran, T., Schultz, P., 2004. Weather, stock returns, and the impact of localized trading behavior. J. Financ. Quant. Anal. 39, 343–364.
Nartea, G.V., Kong, D., Wu, J., 2017. Do extreme returns matter in emerging markets? Evidence from the Chinese stock market. J. Bank. Financ. 76, 189–197.
Nartea, G.V., Wu, J., Liu, H., 2014. Extreme returns in emerging stock markets: evidence of a MAX effect in South Korea. Appl. Financ. Econ. 24, 425–435.
Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55, 703–708.
Ohlson, J.A., 1980. Financial ratios and the probabilistic prediction of bankruptcy. J. Account. Res. 18, 109–131.
Parrott, W.G., Sabini, J., 1990. Mood and memory under natural conditions: evidence for mood incongruent recall. J. Personal. Soc. Psychol. 59, 321–336.
Persinger, M.A., 1975. Lag responses in mood reports to changes in the weather matrix. Int. J. Biometeorol. 19, 108–114.
Persinger, M.A., Levesque, B.F., 1983. Geophysical variables and behavior: the weather matrix accommodates large portions of variance of measured daily mood. Percept. Mot. Skills 57, 868–870.
Petty, R.E., Gleicher, F., Baker, S.M., 1991. Multiple roles for affect in persuasion. In: Forgas, J.P. (Ed.), Emotion and Social Judgments. Pergamon Press,, Oxford, pp. 181–200.
Pirinsky, C.A., Wang, Q., 2006. Does corporate headquarters location matter for stock returns? J. Financ. 61, 1991–2015.
Rapp, A., 2019. Sentiment versus mood: a conceptual and empirical investigation. J. Cap. Mark. Stud. 3, 6–17.
Rind, B., 1996. Effect of beliefs about weather conditions on tipping. J. Appl. Soc. Psychol. 26, 137–147.
Rosenthal, N.E., Sack, D.A., Gillin, J.C., Lewy, A.J., Goodwin, F.K., Davenport, Y., Mueller, P.S., Newsome, D.A., Wehr, T.A., 1984. Seasonal affective disorder: a description of the syndrome and preliminary findings with light therapy. Arch. Gen. Psychiatry 41, 72–80.
Saunders, E.M., 1993. Stock prices and Wall Street weather. Am. Econ. Rev. 83, 1337–1345.
Scholes, M., Williams, J., 1977. Estimating betas from nonsynchronous data. J. Financ. Econ. 5, 309–327.
Schultz, P., 2003. Who makes markets? J. Financ. Mark. 6, 49–72.
Schwarz, N., 1990. Feelings as information: informational and motivational functions of affective states. In: Higgins, E.T., Sorrentino, R.M. (Eds.), Handbook of Motivation and Cognition. Guilford Press,, New York, pp. 527–561.
Schwarz, N., Clore, G.L., 1983. Mood, misattribution, and judgments of well-being: informative and directive functions of affective states. J. Personal. Soc. Psychol. 45, 513–523.
Scott, J., 2007. The impact of weather conditions on mood variability in geographically relocated versus non-relocated individuals. J. Res. Minn. State Univ. 7. Article 12.
Seasholes, M.S., Zhu, N., 2010. Individual investors and local bias. J. Financ. 65, 1989–2011.
Seif, M., Docherty, P., Shamsuddin, A., 2018. Limits to arbitrage and the MAX anomaly in advanced emerging markets. Emerg. Mark. Rev. 36, 95–109.
Sinclair, R.C., Mark, M.M., 1995. The effects of mood state on judgmental accuracy: processing strategy as a mechanism. Cogn. Emot. 9, 417–438.
Stambaugh, R.F., Yu, J., Yuan, Y., 2012. The short of it: investor sentiment and anomalies. J. Financ. Econ. 104, 288–302.
Tversky, A., Kahneman, D., 1992. Advance in prospect theory: cumulative representation of uncertainty. J. Risk Uncertain. 5, 297–323.
Walkshäusl, C., 2014. The MAX effect: European evidence. J. Bank. Financ. 42, 1–10.
Wright, W.F., Bower, G.H., 1992. Mood effects on subjective probability assessment. Organ. Behav. Hum. Decis. Process. 52, 276–291.
Zhong, A., Gray, P., 2016. The MAX effect: an exploration of risk and mispricing explanations. J. Bank. Financ. 65, 76–90.
Zhu, N., 2002. The local bias of individual investors. Unpubl. Work. Pap. Yale Univ.